

TANMAY SOMANI

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Summary

AI/ML engineer with strong foundations in **machine learning, statistics, and data science**. Experienced in **predictive modeling, deep learning, scalable AI systems, automation, and business intelligence solutions**.

Work Experience

AI/ML Research Intern *Indian Space Research Organisation*

Feb 2026 – Aug 2026

- Developed a **ConvLSTM-U-Net** model to forecast **future satellite frames** for short-term weather nowcasting
- Processed **16,500+ INSAT-3DS satellite images**, generating **7-frame input/output sequences** for weather forecasting
- Conducted **model ablation** across architecture, loss functions, and hyperparameters to test spatiotemporal forecasting performance

AI Intern *HR Visko Group*

Aug 2025 – Sep 2025

- Optimized **machine learning pipelines** through model experimentation and inference workflow improvements
- Implemented **automated monitoring and evaluation workflows** for systematic tracking of model performance and experiments
- Analyzed **user interaction datasets** to identify data-driven improvements for prediction quality and downstream decision systems

AI/ML Intern *JS TechAlliance Consulting Private Limited*

Jun 2025 – Jul 2025

- Developed **end-to-end speech AI pipelines** integrating **3 core capabilities**: text-to-speech, transcription, and speaker diarization
- Automated **Python-based conversational audio workflows**, integrating multiple AI components into unified processing pipelines.

Education

School of Data Science and Forecasting, DAVV University

2021 – 2026

Bachelor of Technology – Master of Technology (Dual Degree) in **Artificial Intelligence and Data Science**

CGPA: 8.2 / 10

- Coursework: Probability, Statistics, Machine Learning, Optimization, Linear Algebra
- Certifications in **IoT, Cybersecurity, and Google Cloud**

Technical Skills

Programming:	Python, SQL, R, Scala, Java, C++, Bash, LaTeX
Machine Learning:	Deep Learning, Feature Engineering, Model Evaluation, Time Series Modeling
Libraries:	Pandas, NumPy, PyTorch, TensorFlow, Keras, LangChain
Visualization & Power Platforms:	Power BI, Matplotlib, Power Apps, Power Automate, Power Query
Tools & Platforms:	Git, Docker, FastAPI, MongoDB, MySQL, Linux, GCP, AWS, Azure
Concepts:	Data Analysis, Statistical Modeling, Optimization, Data Pipelines, NLP

Projects

LifeLogAI – Intelligent Memory System

- Built a **multimodal AI memory system** with semantic search and RAG-based retrieval pipelines
- Designed scalable **vector database workflows** for contextual memory retrieval using LLMs

Aetherium – Multi-Agent Workspace

- Developed a **multi-agent AI platform** for automated research and document generation
- Integrated **local LLM pipelines and vector embeddings** for contextual knowledge retrieval

Power BI Sales and Business Intelligence Dashboard

- Designed interactive **Power BI dashboards** for KPI tracking, analytics, and business reporting
- Performed **data cleaning, SQL querying, and visualization** using Power Query and BI tools

Leadership

AIESEC Indore

Aug 2023 – Jul 2024

Head of Outgoing Corporate Sector

- Facilitated international internship opportunities for **100+ students** across Indore
- Managed outreach initiatives and partnerships across multiple international teams

Publications

Comparative Analysis of Reinforcement Learning and Evolution Strategies

Springer Book Chapter

- Published by **Springer**; compared RL and Evolution Strategies across optimization efficiency, reward convergence, and performance.
- DOI: 10.1007/978-3-032-16791-0_14

INSAT-3DS Satellite Weather Forecasting using ConvLSTM-U-Net

ISRO Institutional Journal

- Developed a **ConvLSTM-U-Net** spatiotemporal forecasting pipeline using multi-temporal INSAT-3DS imagery for short-term weather nowcasting.

Certifications & Achievements

- 4th Prize – Indian National Space Hackathon** – BAH2026
- British Airways Data Science Job Simulation** – Forage
- Google Cloud Fundamentals** – GCP

Languages

English (Proficient)

Hindi (Native)